BTC Field Guide

Describing a natural-territorial complex in the field: a short English companion to the BTC Field App

> This guide is an independent, abridged text written for the English interface of the BTC Field App. It is not a translation. It follows the landscape-geophysical field method of Prof. Niko Beruchashvili as taught at Ivane Javakhishvili Tbilisi State University, and explains how to fill in each part of the app's form. The full method, with its theory, examples and tables, is in the Georgian methodology manual (TSU, Tbilisi, 2024), which is the reference wherever the two differ.

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1. Before you start

1.1 What you are describing

A natural-territorial complex (BTC, from its Georgian name) is a piece of land where relief, rock, water, air, soil, plants and animals work together as one system. In the field the smallest such unit, the facies, is the usual object: a patch with one position on the slope, one kind of substrate, one moisture regime and one plant community.

The method looks at a BTC as a vertical column. It runs from the top of the vegetation down through the soil to the rock below, and is made of layers called geohorizons. Each geohorizon is built from geomasses: air, living plant matter, dead plant matter, soil, rock and water. Describing a BTC means recording where it is, what conditions shape it, and how this column is built.

1.2 The three stages of the work

- **Before the field:** collect topographic maps at a large scale, geological and soil maps, satellite images and earlier studies of the area. Decide how detailed the survey has to be.
- **In the field:** choose the plots, describe them on the form, take samples and photographs.
- **After the field:** process samples, compute masses, compare the plots and draw conclusions or maps.

The app supports the field stage. Records can be exported as PDF, Excel, CSV, KML or BTC files for the desk work that follows.

1.3 Choosing the plots

Nobody can observe a whole landscape, so the work is done on a few well chosen plots. The largest is the key plot. Every smaller plot of the same record lies inside it.

- **Key plot:** placed in the most typical BTC, so that what you learn can be carried over to a wider area. Add more key plots where the vertical structure or the state of the BTC clearly changes. In a forest, make it large enough to hold about 50 trees. Lay it out as a rectangle with the long side running down the slope, and measure all four sides with a tape.

- **Sample area:** used to measure the mass of trees and shrubs. Its size depends on the landscape and on how detailed the survey is (see the table below).
- **Record plot:** a small square or rectangle (1 × 1 m, 0.5 × 0.5 m, 1 × 0.5 m or 2 × 1 m) for cutting and weighing grasses and dead plant matter. Usually 3 to 5 per BTC, or a single one in a very simple BTC.
- **Soil pit:** usually 1 m long, 0.5 m wide and 1 m deep. Rock or many stones may stop it at about 0.5 m.

Landscape	Detail	Size (m)	Area (m²)
Complex, sparse or park forests	high	100 × 100	10 000
	medium	100 × 50	5 000
	low	50 × 20	1 000
Medium and low forests	high	100 × 50	5 000
	medium	50 × 20	1 000
	low	25 × 10	250
Semi-arid and semi-humid, with shrubs	high	10 × 5	50
	medium	5 × 2	10
	low	2.5 × 2	5
Semi-desert and desert	high	10 × 5	50
	medium	5 × 2	10
	low	1 × 1	1

A key plot may sit on the border of two neighbouring facies only if both have the same type of vertical structure.

1.4 Naming the record

Give each key plot a short code: an abbreviation of the study area and a number, for example BAK 1. When several teams work in the same area, let the hundreds show the team: BAK 101, 102, 103 for the first team, BAK 201, 202 for the second. Use the code as the record name in the app, and the Group field to keep one expedition or route together.

1.5 Using the app in the field

- **Fill language:** on the Basic tab choose Georgian, English or both. With both, each text field gets a second, English box, and the PDF prints the two versions together.
- **Voice typing:** the app uses the keyboard's own microphone. Gboard on Android recognises Georgian and English. Settings > Tips explains other devices.
- **Offline:** everything is saved on the device first, and synchronised later if you use Cloud or Expedition storage.
- **Photos:** each tab has its own photo panel. Photograph the plot, the soil profile and the vertical structure.

2. Basic tab

2.1 Date, name and group

The date is filled in when the record is created. Change it if you enter notes later. The name is the plot code (section 1.4).

2.2 Location

Describe the place in words so that someone else can find it again without GPS. Tie it to at least two permanent landmarks, such as a village, a bridge, a road junction or a survey point, and give the smallest administrative unit. A position relative to another plot is useful only as an addition.

If the plot lies in the Caucasus, name its landscape after N. Beruchashvili's Landscape Map of the Caucasus (1:1 000 000, 1979) and note the number of the landscape on that map. Using one map for every plot keeps the records comparable.

2.3 Coordinates, altitude and aspect

- **Coordinates:** press Detect via GPS and wait for a good accuracy, or type them in. The converter accepts decimal degrees, degrees and minutes, degrees-minutes-seconds and UTM.
- **Altitude:** height above sea level in metres. Phone GPS height can be tens of metres off, so check it against a topographic map.

- **Aspect:** the direction the slope faces, in degrees from north (0 = north, 90 = east, 180 = south, 270 = west). Leave it empty on flat ground.

Class	Aspect	Degrees
1	North	337.5 - 22.5
2	North-east	22.5 - 67.5
3	East	67.5 - 112.5
4	South-east	112.5 - 157.5
5	South	157.5 - 202.5
6	South-west	202.5 - 247.5
7	West	247.5 - 292.5
8	North-west	292.5 - 337.5

If you recorded a GPS track for the route, link it to the record so that the plot can be placed on the path later.

3. Physico-geographic tab

3.1 Geological formation

Name the rock the BTC is built on, its age if known, and how it lies. The best evidence is a natural outcrop or a cut such as a road cutting or a stream bank. Without one, use a geological map; for Georgia the 1:200 000 sheets are the easiest to obtain. Plants can help: some grow only on limestone, others only on sand.

Write, for example: "Middle Jurassic porphyrite series, weathered at the surface" or "Quaternary alluvium: pebbles and sand".

3.2 Relief type

Look at the group of landforms around the plot, not at a single form, and name the type by the process that made it. Common types are:

- denudational, erosion-denudational, erosional
- erosion-accumulative, accumulative, accumulative-erosional
- karst, arid-denudational
- gravitational, structural (petrogenic)

- glacial and old glacial (paleoglacial)

The relief type says a lot about how stable the landscape is under human use, which is why it is recorded separately from the shape.

3.3 Morphological description

Describe the shape and size of the landform and the plot's place on it: crest, upper, middle or lower slope, foot, terrace, floodplain. Give length and width where they matter, how densely and how deeply the surface is cut by gullies, and the height difference between the lowest and highest point of the plot. State the slope steepness.

Slope (degrees)	Description
0 - 4	flat or almost flat
4 - 10	gentle
10 - 20	slightly steep slope
20 - 30	moderately steep slope
30 - 45	steep slope
45 - 60	very steep slope
60 - 90	cliff, rock face

3.4 Modern geomorphological processes

List the processes active now, and say how strong they are and how much of the plot they affect. The most frequent are:

- weathering debris (scree), rockfall, landslides
- mudflows and debris flows, snow avalanches
- gully erosion and sheet erosion
- nivation, karst and clastokarst
- creep, solifluction and other slow slope movement
- deflation (wind erosion)

3.5 Migration regime

The migration regime describes how water and the matter it carries move through the BTC. It depends on the position in the relief and on the slope.

- **Eluvial (autonomous):** flat or almost flat watersheds. Water comes only from precipitation; matter is leached downwards and out.
- **Transeluvial:** slopes. Matter both arrives from above and leaves downslope, so the BTC is a place of transit.
- **Eluvial-accumulative and accumulative-eluvial:** lower slopes and foot slopes, where matter starts to accumulate.
- **Transaccumulative:** hollows and valley floors with throughflow.
- **Superaquatic:** groundwater is close to the surface and reaches the soil.
- **Subaquatic:** under water, at the bottom of lakes and rivers.

When the vertical structure index is written (section 7.4), the regime and slope together give a number at its start:

No.	Regime and slope
1	autonomous, 0 - 4°
2	autonomous, 4 - 10°
3	transeluvial, 10 - 20°
4	transeluvial, 20 - 30°
5	transeluvial, 30 - 45°
6	transeluvial, over 45°
7	badlands
8	cliffs
9	eluvial-accumulative, 10 - 20°
10	eluvial-accumulative, 4 - 10°
11	eluvial-accumulative, 0 - 4°
12	superaquatic, 0 - 4°
13	superaquatic, 4 - 10°
14	transaccumulative
15	accumulative-eluvial

Do not apply the slope classes mechanically. On the upper and middle slope the transeluvial regime should prevail, on the lower slope the accumulative one. Look also at the relief and regime of the landscape above the plot.

3.6 Moisture degree

Record where the moisture comes from and how much there is. Most places have more than one source, and the source can change with the state of the BTC during the year.

- **Atmospheric:** moisture from precipitation only.
- **Ground:** from groundwater, with or without springs on the plot.
- **Deluvial:** from water running down the surface of the slope.
- **Floodplain:** from floods.

Degree	Soil	Typical state
insufficient	very dry	arid
weak	partly moist	semi-arid
optimal	moist	humid
abundant	very moist	humid
excessive	wet	extra-humid

Note the groundwater level if you can find it, from the water in a well or from the depth at which water seeps into the soil pit. Willow, alder, reeds and sedges also point to shallow groundwater.

4. Vegetation tab

4.1 What to record

Work inside the key plot. First list every species you find. If you cannot name a plant, give it a temporary name or number and take a specimen for the herbarium, so that it can be identified later.

Enter one row per tier and species group. Each row has a tier, a height, a density, a phenophase and the species.

4.2 Tier

Each tier is a fixed height band, so the classes are the same in every landscape and records can be compared.

Tier	Height	Tier	Height
I	0 - 5 cm	VI	2 - 4 m
II	5 - 25 cm	VII	4 - 8 m
III	25 - 50 cm	VIII	8 - 16 m
IV	50 - 100 cm	IX	16 - 32 m
V	1 - 2 m	X	over 32 m

> The tiers in this table grow with height: tier I is the lowest layer. Check the convention used by your course before you start, and use the same one for every record of the survey.

4.3 Height

Give the average height of the species or layer in metres. Measure grasses together with their flowers and seed heads. Trees can be measured with a clinometer or estimated against a pole of known length.

4.4 Density

Density is the share of the ground that the plants cover when seen from above (projective cover). Use one of two scales and say which.

Drude	Meaning
sol.	very rare, almost no cover
+	rare, small cover
1	plentiful with very little cover, or rare with large cover
2	quite plentiful, cover under 1/20 of the area
3	cover 1/4 to 1/2 of the area
4	cover 1/2 to 3/4 of the area
5	cover over 3/4 of the area

The ten-point scale gives one point for each 10 % of cover: 10 points is 90 - 100 %, 5 points is 40 - 50 %, 1 point is under 10 %. Note also whether the plants grow evenly, in clumps or in patches.

4.5 Phenophase

Code	Phase	Code	Phase
1	vegetative growth	5	seed ripening and shedding
2	budding	6	flowers and leaves falling
3	flowering	7	drying, dying back
4	fruiting	8	dormancy

4.6 Naming the community

Name a plant association after the dominant species of its main tier, with the most important species, the dominant or edificator, last: a hornbeam-oak forest is an oak forest with hornbeam. A plant formation is named after the dominants of all tiers, for example "hornbeam-oak forest with deciduous shrubs".

In forests also note the regrowth: how many young trees there are, which species, and whether they are spread evenly. Uneven regrowth needs an explanation, such as grazing or a gap in the canopy.

5. Soil tab

5.1 The soil pit

Dig the pit where the soil is typical for the plot, with the face to be described in good light. Fix a measuring tape to the face with zero at the surface, mark the horizon boundaries with a knife, and describe each horizon from the top down.

5.2 Soil type name

Give the type, subtype and variety, for example "brown forest soil, leached, medium loam, shallow". Note any sign of human change: ploughing, terracing, compaction, erosion.

5.3 Genetic horizons

Add one row per horizon. Write its symbol and depth, for example A1 0-12 for an A1 horizon from the surface to 12 cm. The usual symbols are:

- **A0:** litter, dead plant material on the surface.
- **A1:** dark humus-accumulative horizon.
- **A2:** pale eluvial horizon, leached of humus and clay.
- **Ap:** ploughed layer.
- **A/B or A3:** transition from A to B.
- **B:** illuvial horizon, usually denser and browner, where substances from above accumulate. B1, B2, B3 by decreasing humus colour; Bk with accumulated carbonates.
- **G:** gley horizon, grey or bluish-green, formed under constant waterlogging, often with rusty spots.
- **C:** parent material not changed by soil formation.
- **D:** underlying rock of a different kind below the parent material.

Small letters after the symbol add detail: g signs of gleying, t clay accumulation, h illuvial humus, k or ca carbonates, cs gypsum, sa other soluble salts, sn concretions, m a cemented layer.

5.4 Horizon description

The app's legend lists the properties in a fixed order. Describe each that you can see:

- **Colour:** darker means more humus; brown and red tones come from iron and manganese; whitish tones from leaching or carbonates.
- **Texture (mechanical composition):** clay, heavy, medium or light loam, sandy loam, sand. Judge it by rolling moist soil between the fingers.
- **Structure:** the shape of the aggregates: crumb, granular, nutty, blocky, prismatic, platy, or structureless.
- **Porosity:** fine, medium or coarse pores; cracks.
- **Density:** loose, slightly compact, compact, very compact (how hard it is to push in a knife).
- **New formations:** carbonate spots and veins, iron and manganese concretions, rust mottles, salt crystals.
- **Gravel % and roots %:** share of stones and of roots in the horizon.
- **Moisture:** dry, slightly moist, moist, wet.
- **Humus type:** mull, moder or mor.
- **Lower boundary:** sharp (transition 2 - 3 cm), clear (3 - 5 cm) or gradual (over 5 cm); also its shape: even, wavy, tongued.

5.5 Geohorizon index and surface formation

For each soil horizon write the geohorizon it belongs to (chapter 7). Under soil surface formation describe what covers the soil: litter, moss, stones, crust, bare ground.

6. Geomass tab

6.1 Geomasses

A geomass is a body of matter of one kind inside the BTC. Its class is written with a capital letter, and smaller letters add the type. The main classes are:

Index	Geomass	Examples
A	aeromass	air between and above the plants
P	phytomass	living plants: Pt trunks and branches, Pf leaves, Ps roots
M	mortmass	dead plant matter: standing dead grass, litter, fallen wood
S	pedomass	fine soil, by texture: Sa clay, Ss loam, Sc sandy loam, Sp sand
L	lithomass	stones and rock: Ls silicate, Lk carbonate
H	hydromass	water: H' too little, H'' optimal, H''' too much

The Georgian manual gives the full list of types, for example leaves of mesophytes, xerophytes and evergreens, or needles and grass stems.

6.2 Units

Masses are given per area, usually in tonnes per hectare (t/ha) or grams per square metre (1 t/ha = 100 g/m²). Bulk density and density are given in g/cm³.

6.3 The geomass table

Enter one row per layer of the soil pit, with its depth in cm.

- **Pedomass:** bulk density (g/cm³) and quantity (t/ha). Quantity = bulk density × layer thickness (cm) × share of fine soil in the layer (as a fraction) × 100. Example: 1.0 g/cm³, 100 cm, 50 % gives 5 000 t/ha.
- **Lithomass:** density of the rock (g/cm³) and quantity (t/ha), computed the same way with the share of stones.
- **Hydromass:** soil moisture wet and dry, and the quantity of water. Quantity = pedomass of the layer × moisture (%) / 100.
- **Phytomass:** wet and dry mass of roots in the layer.

Add the layers to get the total for the profile. In mountains the soil is rarely deeper than 1 m, so totals usually cover the top metre.

Typical rock densities (g/cm³)

Rock	Density	Rock	Density
granite, gneiss	2.7	sandstone, clay	2.3
basalt	2.8	conglomerate	2.5
porphyrite, andesite	2.7	limestone	2.5
shale	2.7	marl	2.6
heavy tuff	2.4	weathered rock	1.5 - 2.5

Soil moisture by weighing

Take samples at fixed depths (for example 0-5, 5-10, 15-20, 25-30, 35-40, 45-50 cm and deeper), seal each in a bag with a label giving the plot, date and depth, and weigh them in closed tins. Dry them at 100 - 105 ° C for 5 - 7 hours, weigh, dry for another hour and weigh again, until the weight stops changing. Pick out and weigh gravel over 3 mm. Then:

moisture (%) = 100 × (wet weight - dry weight) / (dry weight - tin - gravel), where the wet and dry weights include the tin.

6.4 Tree phytomass

In the sample area count the trees, measure the diameter at breast height and the height of each, and find the stem volume from volume tables. Enter the plot size, and for each species the volume and the wet and dry mass of wood, leaves, branches and roots in t/ha. The app adds the totals.

7. Vertical structure tab

7.1 The sketch

Start with a coloured drawing of the BTC column, from the tree tops to the rock. Give each geomass its own colour, for example brown for trunks and branches, dark green for needles and light green for leaves and grass. In forests use two scales: one for the whole column and a larger one for the thin layers near the ground. Attach a photo of the sketch on this tab.

7.2 Separating geohorizons

Go down the column. A new geohorizon starts where a geomass appears or disappears, where the ratio between geomasses changes clearly, or where their shape and arrangement change, for example where the crown gives way to bare trunks. Separate geohorizons of the same rank throughout the profile. Record for each one which soil horizon or plant tier it corresponds to.

7.3 Geohorizon index

- **Order:** write the geomass classes from the heaviest to the lightest, with no comma between classes. AP is a layer of air with plants; PA is a layer where plants weigh more than the air.
- **Cover:** after each geomass give its cover in % (plants, air) or its share of the volume (soil, stones). Air and water get no number.
- **Limits:** end the index with the upper and lower limits in metres.
- **Leave out minor parts:** include a geomass only if it is over 1 % of the weight or over 5 % of the cover or volume, and mention the rest in the description.
- **Repeat nothing:** a property shared by several geohorizons is written only once, in the highest one.

Some rules of thumb: forest crowns are mostly air (A first), the trunk zone below them mostly plant (P first). Under grass the class changes when plant cover reaches 70 - 90 %. Stones are two to three times denser than soil, so they come first when they fill over 30 % of a layer. Trunks and branches outweigh leaves even when they cover less, so Pt is written before Pf.

7.4 Vertical structure type

The class of the structure is named after the geomass that shapes it: phytogenic (living plants), postphytogenic (dead plant matter), hydrogenic (snow, frozen ground, standing water, ice), pedogenic (bare soil, such as a harvested field) or lithogenic (rock, beach). Transitional classes are common. The type is then written as a capital letter:

Code	Type
K	Colchic humid forests and shrubs
F	humid mesophytic broadleaf forests (beech, lime, hornbeam, maple, elm)
SA	semi-arid mesophytic forests (oak, oriental hornbeam)
T	humid dark coniferous forests (spruce, fir)
Q	semi-arid light coniferous forests (pine)
H	extra-humid hydrophytic forests (alder, willow, poplar)
TF	humid mixed broadleaf and dark coniferous forests
P	xerophytic open woodland (arid open forests, juniper)
L	humid high-mountain grassland (alpine and subalpine meadows)
SH	semi-humid and semi-arid grassland (meadow steppe, steppe)
G	humid meadows of low and middle mountains
A	arid deserts and semi-deserts
C	cliffs

Marshes, glaciers, farmland and settlements have their own signs, which are given in the Georgian manual.

7.5 Height, thickness and complexity

Enter the total height of the structure in metres. Its thickness is the distance from the upper to the lower limit, without the air above and the rock below, and is graded on a doubling scale:

Code	Thickness grade	Complexity	Geohorizons
0	nano, under 1 m	primitive	2 - 3
1	micro, 1 - 2 m	simple	4 - 5
2	meso, 2 - 4 m	medium	6 - 7

Code	Thickness grade	Complexity	Geohorizons
3	mesomacro, 4 - 8 m	complicated	8 - 9
4	macromeso, 8 - 16 m	complex	10 - 11
5	macro, 16 - 32 m	very complex	over 12
6	mega, over 32 m		

Intensity is the number of geohorizons per metre of profile: low (one per 2 m), medium (one per 1 - 2 m) or high (one per metre).

7.6 Writing the full index

The landscape unit index joins these parts: the migration regime number, the structure type, small letters for the substrate and additions, the thickness grade, and the landscape number from the map in brackets. For example 5Fk5w (71): a transeluvial slope of 30 - 45° (5), humid beech forest (F) on carbonate rock (k), structure 16 - 32 m thick (5), with evergreen Colchic shrubs (w), landscape 71.

- **Additions:** n dead cover, i low grass, ii mixed herbs, iii tall herbs, w evergreen shrubs, v deciduous shrubs, x xerophytes, p ferns, g halophytes, m mosses and lichens, d young and secondary forest.
- **Substrate:** c crystalline, t terrigenous, k carbonate, s saline, a alluvial, p proluvial, g glacial, o colluvial deposits.

7.7 Description

In the description field, explain how the part above ground relates to the part below: how deep the roots reach compared with the height of the plants, where water and litter collect, and how the layers depend on each other. Mention the geomasses left out of the indexes.

8. Glossary

- **BTC:** natural-territorial complex, the unit of landscape study.
- **Facies:** the smallest BTC, uniform in position, substrate, moisture and plant cover.
- **Geomass:** a body of matter of one kind (air, plants, dead plant matter, soil, rock, water) inside a BTC.
- **Geohorizon:** a layer of the BTC column with a uniform set and ratio of geomasses.
- **Vertical structure:** the order of geohorizons and the links between them.
- **State (stex):** the condition of a BTC over one day; a BTC passes through several states in a year.
- **Key plot:** the main plot where a BTC is described.
- **Migration regime:** how water and matter move into, through and out of a BTC.
- **Projective cover:** the share of the ground covered by plants, seen from above.
- **Mortmass:** dead plant matter, standing or lying.
- **Edificator:** the species that shapes the community and its environment.

Sources

N. Beruchashvili. Methods of mapping the states of natural-territorial complexes and of landscape-geophysical research. Tbilisi, 1983 (in Russian).

Methodology of landscape field research and mapping, compiled for geography students of Ivane Javakhishvili Tbilisi State University, Tbilisi, 2024 (in Georgian; translated from the 1983 manual by D. Nikolaishvili, with additions from T. Gordeziani, 2014). This is the full Georgian methodology bundled with the app.

N. Beruchashvili. Landscape Map of the Caucasus, 1:1 000 000. Tbilisi, 1979.